[COMPANY NAME REDACTED]

the Platform Software Requirements Specification

Requirements for freight operations, operation approval, billing, accounting bridge, general ledger and tracking

Document	Software Requirements Specification (SRS)
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Prepared by	Engineering
Owner	the Company management

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Companion document: System Analysis and Design (SAD) v0.1, which describes how these requirements are realised. Priority: **Must** = required for release; **Should** = required unless the owner accepts a stated alternative; **Could** = desirable. Status: **Built**, **Partial**, **Planned** (see Section 1.5).

Document Control

Revision History

Version	Date	Author	Description
0.1	November 8, 2026	Engineering	First issue, based on a stale local checkout.
0.1 (rev. 2)	November 8, 2026	Engineering	Re-based on origin/main. Adds the operation approval requirements, air consolidation, the bridge and ledger states, the public lookup with proof of work, the demo seed, and the known gaps from the demo rehearsal as Planned or limitation items.

Review and Approval

Role	Responsibility	Signature	Date
Business owner	Confirms the requirements and priorities		
Accounting lead	Confirms ledger, billing, bridge and period rules		
Operations lead	Confirms freight, approval, cartage, dispatch and tracking requirements		
Technical lead	Confirms feasibility, interfaces and non-functional targets		

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1. Introduction

1.1 Purpose

This specification states the requirements of the Platform, the internal operations and accounting platform of the Company. It is the reference for design, build, test and acceptance. Each requirement has an identifier, a priority, a status and acceptance criteria, so that delivery can be checked against it.

1.2 Scope

The platform covers freight files for ocean and air movements, including air consolidation; the operation approval of files; vehicle inventory for roll-on/roll-off shipments; pickup and delivery cartage; fleet, drivers and dispatch; rates, tariffs, quotes and commodities; billing lines and invoices; the accounting bridge; the general ledger with vouchers, journals, period close and reports; tracking, the public lookup and the live map; and administration of users, companies, audit records, attachments, support requests and demonstration data.

The platform does not cover payroll, customer relationship management, bank file transfers, or direct submission to government systems.

1.3 Intended audience

- Business owner and management, who approve scope and priorities.
- Accounting staff, who confirm the billing, bridge and ledger rules.
- Operations staff, who confirm freight, approval, cartage, dispatch and tracking rules.
- Engineers and testers, who build and verify the platform.

1.4 Definitions and acronyms

The glossary in the companion SAD (Section 16) defines the terms used here.

Table 1-1. Acronyms

Acronym	Meaning
AP / AR	Accounts payable (costs to suppliers) / accounts receivable (charges to customers)
CDV	Check disbursement voucher
FR / NFR / IFR / DR / FR-AP	Functional, non-functional, interface and data requirement; FR-AP is the operation approval group
FS	The general ledger module
HAWB / MAWB	House air waybill / master air waybill
JWT	JSON Web Token, used for sign-in tokens
POD	Proof of delivery
RBAC	Role-based access control
SAD / SRS	System analysis and design / software requirements specification
VAT	Value-added tax (12 percent on taxable billing lines)

1.5 Status labels and sources

Table 1-2. Status labels

Label	Meaning

Label	Meaning
Built	Implemented in the repository at origin/main. The acceptance column names the check or review that proves it.
Partial	Implemented in part. The missing part is named in the requirement.
Planned	Not implemented in the repository. Listed in Appendix A.

Sources: the project README, the demo known-gaps list, the demo script and demo data notes, the freight standards notes, the database schema, the route and library source, the continuous integration workflow and the example configuration file. Configuration values, credentials and customer records are not reproduced.

2. Overall Description

2.1 Product perspective

the Platform is a web application with one sign-in. It replaces spreadsheet and paper steps in freight operations, adds an approval step before billing, and links billing to the ledger through an accounting bridge. The public lookup is the only part that does not need a sign-in.

2.2 Product functions

1. Sign in, sign out, session control and access by role and company.
2. Show an operations overview and a workspace for each role.
3. Book ocean and air files, including air consolidation under a master air waybill.
4. Approve freight files: submit, approve, return with a note, and lock.
5. Maintain rates, tariffs, charge packages, quotes and commodities.
6. Rate approved files into billing lines, drafts, invoices, voids and reversals.
7. Stage, trial post and post billing and cost items to the ledger.
8. Keep the company ledger: vouchers, journals, chart of accounts, period close and reports.
9. Manage vehicles, cartage orders, fleet and dispatch with proof of delivery.
10. Record milestones; provide a public lookup; show branches, routes and live positions on a map.
11. Administer users, companies, audit records, attachments, integration status, support requests and demonstration data.

2.3 User classes and characteristics

Table 2-1. User classes

Class	Characteristics	Principal functions
superadmin	Technical administrator; full access; only role that can change another superadmin account	All functions, including approvals and administration
manager	Supervises operations; approves and returns files; no user administration	Files, approvals, logistics and ledger functions
operator	Encoder: creates and edits draft and returned files and submits them; cannot approve; no ledger access	Files, vehicles, cartage, fleet, dispatch, tracking
accountant	Owns billing review, the bridge and the ledger; sees approved files only	Billing, bridge, ledger, reports
viewer	Reads records; cannot write	Read access to modules
Public visitor	Any person with a reference number; no account	Public lookup

2.4 Operating environment

Table 2-2. Operating environment

Element	Environment
Client	Current desktop and mobile browsers with JavaScript; layouts for 390-pixel screens and wider desktops; light and dark appearance
Server	Node.js 22 or later at run time; Express API; Prisma client for the application database

Element	Environment
Databases	Two SQLite files: application database and accounting database
Hosting	Linux-based Azure App Service, production slot, with persistent storage and application settings for secrets
Build and deploy	GitHub Actions; checks on pull requests; deployment on push to main

2.5 Constraints

- **C-01** Production has no migration command. Schema changes must be additive and applied at start-up.
- **C-02** Secrets are held only in host settings; never committed or sent to the browser.
- **C-03** Production records must not be used for testing or demonstrations.
- **C-04** Ledger postings are made only through the accounting bridge.
- **C-05** Amounts are held with their currency (PHP, USD, EUR, JPY, CNY and others); the ledger records in PHP.

2.6 Assumptions and dependencies

- **A-01** Tariffs and charge rates are entered by the business. Most lanes have no tariff yet; a lane without one shows "no tariff" until added.
- **A-02** The ledger chart contains the accounts that billing codes map to.
- **A-03** Public data services for ship and aircraft positions, exchange rates and vehicle lookup remain available under their own terms; the platform degrades when they are not.
- **A-04** The owner rotates development credentials before go-live.

3. External Interface Requirements

3.1 User interface

ID	Requirement	Priority	Status	Acceptance
IFR-01	The system shall present one shell with navigation to the logistics suite and the ledger, and show the signed-in company and user.	Must	**Built**	Screen check at desktop and 390 px width
IFR-02	Every list screen shall show loading, empty and error states.	Should	**Partial**	Screen check of each list
IFR-03	Money values shall be shown with tabular figures and the document currency.	Should	**Built**	Visual check on billing and ledger screens

3.2 Software interfaces

ID	Requirement	Priority	Status	Acceptance
IFR-04	The API shall be JSON over HTTPS under /api, with bearer tokens for every route except sign-in, challenge, public lookup and support.	Must	**Built**	Route tests refuse calls without a token
IFR-05	The API shall return a consistent error body with a message the user can read.	Should	**Partial**	Negative tests on each route group
IFR-06	The server shall call public ship, aircraft, exchange-rate and vehicle-lookup services only through configured endpoints; an unset aircraft endpoint reports not configured.	Must	**Built**	Configuration review; feed states tested
IFR-07	Device connectors (barcode scanner, signature pad, label printer), government customs filing and carrier rate APIs shall report connected, disconnected or error.	Could	**Partial**	Status records exist; live connectors are Planned (FR-AD-05)

3.3 Communications interfaces

ID	Requirement	Priority	Status	Acceptance
IFR-08	All browser traffic shall use HTTPS in production.	Must	**Built**	Hosting configuration review
IFR-09	Request bodies shall be limited to 1 MB JSON, 100 KB form and 15 MB on the attachment route.	Must	**Built**	Oversize request is refused
IFR-10	Browser connections shall be limited by the content security policy to the application, the map tile hosts and the public data hosts.	Must	**Built**	Header inspection on a deployed build

4. Functional Requirements

Each requirement uses the form "The system shall ...". Acceptance criteria state the observable result that proves it. Where a named check in the continuous integration pipeline proves it, the check is named.

4.1 Access, accounts and sessions (FR-AU)

ID	Requirement	Priority	Status	Acceptance
FR-AU-01	The system shall let a user sign in with a username and password and return an access token and a refresh token.	Must	**Built**	A valid sign-in returns both tokens and the user
FR-AU-02	The system shall renew the access token with the refresh token while it is valid, without a new sign-in.	Must	**Built**	Expired access token renews in the browser
FR-AU-03	The system shall revoke the refresh token on sign-out.	Must	**Built**	A revoked token is refused
FR-AU-04	The system shall block sign-in for one minute after five consecutive failed attempts.	Must	**Built**	Sixth attempt within the minute is refused
FR-AU-05	The system shall require a solved, signed, expiring, single-use proof-of-work token before a sign-in is accepted.	Must	**Built**	check:captcha and check:captcha-enforcement pass; replay refused
FR-AU-06	The system shall enforce the password policy, including a 72-byte maximum.	Must	**Built**	check:security passes
FR-AU-07	The system shall keep a password history and refuse reuse of a recent password on change.	Must	**Built**	Reuse of a recent password is refused
FR-AU-08	The browser shall end the session after a period of inactivity.	Should	**Built**	Session ends after the idle time
FR-AU-09	The system shall support five roles: superadmin, manager, operator, accountant and viewer.	Must	**Built**	Role list matches Table 2-1
FR-AU-10	The server shall check the role of the signed-in user on every protected route.	Must	**Built**	A role without permission gets a refusal from the API
FR-AU-11	The server shall refuse every write request from the viewer role.	Must	**Built**	Viewer write attempts return a refusal
FR-AU-12	Only a superadmin shall change a superadmin account; a manager shall not manage superadmin accounts.	Must	**Built**	check:security user policy passes

4.2 Operations overview (FR-OV)

ID	Requirement	Priority	Status	Acceptance
FR-OV-01	The system shall show after sign-in a draggable overview with counts of files, vehicles on hold, open cartage, active dispatches and staged bridge items, a to-do list, and each count opening its module.	Should	**Built**	Counts match module lists; no sample numbers shown
FR-OV-02	The system shall show a workspace for each role.	Should	**Built**	Each role sees its own workspace

4.3 Operation approval (FR-AP)

ID	Requirement	Priority	Status	Acceptance
FR-AP-01	The system shall let operator, manager and superadmin users create files in draft and edit them while they are draft or for correction; other roles shall not create files.	Must	**Built**	Viewer and accountant create attempts refused
FR-AP-02	The system shall move a draft or returned file to pending on submit, after checking the required fields (shipper, consignee, bill-to, origin, destination, dates, cargo), and record the revision and the submitter.	Must	**Built**	Incomplete file cannot submit; event written
FR-AP-03	The system shall lock a pending file for every user, including its submitter, until it is approved or returned.	Must	**Built**	Edit attempt on pending refused
FR-AP-04	The system shall let a manager or superadmin approve a pending file, moving it to approved and recording the reviewer.	Must	**Built**	Approval event with user and time
FR-AP-05	The system shall let a manager or superadmin return a pending file with a required note and an optional image, moving it to for correction; a resubmit returns it to pending.	Must	**Built**	Return without a note refused; note and image shown to the operator
FR-AP-06	The system shall lock an approved file for operators, let accountants see approved files only, and refuse any generic write to approval fields.	Must	**Built**	check:opstatus passes; accountant query returns approved files only
FR-AP-07	The system shall prevent a user from approving a file that the same user submitted.	Should	**Planned**	Decision pending (risk R-02); acceptance to follow the decision

4.4 Freight files (FR-SH)

ID	Requirement	Priority	Status	Acceptance
FR-SH-01	The system shall create ocean and air files for export, import and domestic direction, with a file number generated by the system.	Must	**Built**	New file receives a unique number
FR-SH-02	The system shall record shipper, consignee, notify party, ports or airports, carrier, vessel or flight and voyage.	Must	**Built**	Saved values reload unchanged
FR-SH-03	The system shall record containers (type, number, seal) and cargo dimension lines.	Must	**Built**	Lines kept in order after save
FR-SH-04	For ocean files the system shall compute revenue weight as the larger of gross tonnes and cubic metres.	Must	**Built**	check:billing passes
FR-SH-05	For air files the system shall compute chargeable weight as the larger of gross kilograms and volumetric weight (length times width times height divided by 6000 per piece), rounded by the carrier rule.	Must	**Built**	check:air-booking passes for sample dimensions
FR-SH-06	The system shall apply 12 percent VAT to taxable billing lines, and each invoice shall carry VAT treatment 12 percent, zero or exempt.	Must	**Built**	check:invoice passes
FR-SH-07	The system shall let an accepted quote be linked to a file so that its lines replace tariff sell lines.	Should	**Built**	check:billing: quote lines replace sell lines
FR-SH-08	The form shall suggest ports and airports from the party country, default the currency to USD with 20 or more currencies, prefill commodities, and warn on duplicate booking and master numbers.	Should	**Built**	check:air-forms and check:validation pass
FR-SH-09	The system shall print bill of lading or air waybill, packing list, commercial invoice, manifest, export declaration and arrival notice from the file, with branded layouts.	Should	**Built**	Documents print from the file; layout checks in the test suite
FR-SH-10	The system shall let one master air waybill (MAWB) carry many house air waybills (HAWB), entered as rows on one screen (for example 5 or 40 rows).	Must	**Built**	MAWB with five HAWB rows saves and reloads
FR-SH-11	The system shall roll up pieces and gross weight of HAWB rows to the MAWB, and set the MAWB chargeable weight to the greater of total gross and total volumetric weight.	Must	**Built**	check:air-booking roll-up cases pass
FR-SH-12	The system shall produce the consolidation manifest, MAWB, HAWB and an arrival notice per HAWB from the rows; duplicate HAWB numbers shall be flagged.	Must	**Built**	Documents read the rows; duplicate flagged

4.5 Rates, tariffs, quotes and commodities (FR-RT)

ID	Requirement	Priority	Status	Acceptance
FR-RT-01	The system shall maintain directories of carriers, ports and airports, airlines, cargo warehouses and parties (customers, shippers, consignees, agents, truckers, consolidators) with search.	Must	**Built**	Create, search and edit each directory
FR-RT-02	The system shall maintain billing codes with default rate, ledger account and VAT flag.	Must	**Built**	Code saved with all fields; used by billing
FR-RT-03	The system shall apply the most specific tariff (customer and lane, then lane, then any) and ignore expired or future tariffs.	Must	**Built**	check:billing specificity and validity cases
FR-RT-04	The system shall maintain quotes with status Draft, Sent, Accepted, Rejected or Expired, and convert an accepted quote into a file.	Should	**Built**	Status changes and conversion work
FR-RT-05	The system shall maintain a commodity catalogue with HS code and a dangerous-goods flag, and let a free-text commodity be added to it from a booking.	Should	**Built**	Commodity saved with HS code; DG flag shown

4.6 Billing and invoices (FR-BL)

ID	Requirement	Priority	Status	Acceptance
FR-BL-01	When an approved file is saved, the system shall generate billing lines (sell) and cost lines (buy) from tariffs, charge packages or the accepted quote, each with its tariff reference.	Must	**Built**	check:billing passes
FR-BL-02	The system shall never change a manual billing line when generated lines are recalculated.	Must	**Built**	Manual values unchanged after recalculation
FR-BL-03	The system shall raise a sell amount to the tariff minimum when the calculated amount is lower.	Must	**Built**	check:billing minimum case
FR-BL-04	The system shall create an invoice draft per file with a reference (mode code, date, daily sequence) and a revision number; one live invoice per file.	Must	**Built**	check:invoice: reference format and uniqueness
FR-BL-05	The system shall issue an invoice with an invoice number (mode code, year and month, monthly sequence), fixed once issued, and show the FS journal preview before posting.	Must	**Built**	Issue assigns the number once; preview matches posting
FR-BL-06	The system shall void a posted invoice by creating a reversal voucher in the open period, and draft the next revision from the file; posted data shall not be edited.	Must	**Built**	check:flow void case; reversal voucher present
FR-BL-07	The system shall show a billing pipeline listing files by billing stage, and a file view of billing flow.	Should	**Built**	Pipeline counts match the file list
FR-BL-08	The system shall support a company billing-to-ledger mode: manual (drafts only), review (default; staged for one-click posting), or auto (posts when the period is open, vouchers balance and no rate note is open).	Must	**Built**	check:flow runs each mode
FR-BL-09	The system shall issue a credit note document for an invoice with a zero or negative total.	Must	**Planned**	Known gap 1; today such an invoice is refused and corrected by void and re-issue
FR-BL-10	The system shall post charges to revenue accounts by service (freight, handling, documentation) rather than to one sales account.	Should	**Planned**	Known gap 2; waits for the accountants' decision on the mapping

4.7 Vehicles, roll-on/roll-off (FR-VH)

ID	Requirement	Priority	Status	Acceptance
FR-VH-01	The system shall record each vehicle with identification number, year, make, model, colour, shipper, consignee, ports, booking number, location and condition.	Must	**Built**	All fields save and reload
FR-VH-02	The system shall move a vehicle through Expected, Received, Ready to Ship, Pre-Loaded, Loaded, Shipped or Hold, recording each change in history.	Must	**Built**	Each change appears in history
FR-VH-03	The system shall record title status (Pending, Received, Missing), customs hold and lien release flags, and may decode a US-format VIN through a public service.	Must	**Built**	Flags filter the list; decode shows the engine layout label (limitation)

4.8 Pickup and delivery cartage (FR-PD)

ID	Requirement	Priority	Status	Acceptance
FR-PD-01	The system shall create cartage orders of type Pickup, Delivery, cross-dock or exchange, with an order number, addresses, schedule and cargo items.	Must	**Built**	Order saves with a unique number
FR-PD-02	The system shall assign a driver and equipment to a cartage order.	Must	**Built**	Assignment shows on order and driver
FR-PD-03	The system shall move a cartage order through Open, Assigned, In Transit, Delivered or Cancelled, and link it to a freight file.	Must	**Built**	Status and link visible on both records

4.9 Fleet, drivers and dispatch (FR-FD)

ID	Requirement	Priority	Status	Acceptance
FR-FD-01	The system shall maintain a fleet registry with status Available, In Use or Maintenance.	Must	**Built**	Status filter works
FR-FD-02	The system shall maintain a driver registry with status Available, On Trip or Off Duty.	Must	**Built**	Status filter works
FR-FD-03	The system shall create dispatch routes with driver, vehicle, stops and cargo reference, and move each through Draft, Dispatched, In Transit, Arrived and Completed.	Must	**Built**	Route moves through every stage
FR-FD-04	The system shall capture a proof-of-delivery signature with signer and time, only when the signature device connector is connected.	Should	**Partial**	Capture disabled when connector not connected; connector Planned

4.10 Accounting bridge (FR-BR)

ID	Requirement	Priority	Status	Acceptance
FR-BR-01	The system shall stage AR invoice items and AP cost items from billing as Staged, once each.	Must	**Built**	Staging twice creates one item
FR-BR-02	The system shall show staged items in an operations queue with a stage funnel and a drawer for detail.	Must	**Built**	Queue lists items with status
FR-BR-03	The system shall map each charge code to a ledger account and refuse an item whose code is unmapped, giving the reason.	Must	**Built**	check:bridge: unmapped code refused
FR-BR-04	The system shall run a trial post that builds the journal and checks debits equal credits without writing, then set Trial Verified.	Must	**Built**	Trial leaves the ledger unchanged
FR-BR-05	The system shall post a trial-verified item as a balanced journal, set Posted and record the journal reference.	Must	**Built**	Posted journal balances; reference in audit report
FR-BR-06	The system shall keep a check disbursement register with Draft, Printed, Posted and Void, and record the journal reference of a posted check.	Should	**Built**	Each status change recorded
FR-BR-07	The system shall fix voucher numbers per invoice (sales invoice, receipt, reversal) and check them against the ledger before every write.	Must	**Built**	Repeat run writes no duplicate voucher

4.11 General ledger (FR-FS)

ID	Requirement	Priority	Status	Acceptance
FR-FS-01	The system shall maintain the chart of accounts as a tree of account, sub-account and subsidiary, with group codes and subsidiary groups.	Must	**Built**	Tree renders and saves at each level
FR-FS-02	The system shall record cash disbursement vouchers with bank, supplier and signatory data.	Must	**Built**	Voucher saves and prints
FR-FS-03	The system shall record journal entries of types CDV, JE, AJE, AR and AP, each balanced before saving.	Must	**Built**	Unbalanced entry refused
FR-FS-04	The system shall refuse a new or re-dated entry in a closed period and name the first open date.	Must	**Built**	Entry in closed month refused with message
FR-FS-05	The system shall post entries to the ledger and show posted entries in the register for the open period.	Must	**Built**	Posted entry appears in the register
FR-FS-06	The system shall run month-end close for the open period, create closing entries and mark the period closed.	Must	**Built**	Period closed; closing entries exist
FR-FS-07	The system shall produce the trial balance, income statement, balance sheet, cash flow, check voucher report and subsidiary schedule.	Must	**Built**	Totals agree with posted journals (rehearsal reconciliation)
FR-FS-08	The system shall provide a query browser for ledger transactions, including closed periods.	Should	**Built**	Query returns closed-period entries
FR-FS-09	The system shall define the chart and posting accounts for the company of this deployment only, and shall provide no ledger data to companies without a chart.	Must	**Built**	Company without chart returns no ledger accounts

4.12 Tracking and customer accounts (FR-TR)

ID	Requirement	Priority	Status	Acceptance
FR-TR-01	The system shall create a tracking item linked to a freight file, a cartage order or a vehicle, with a reference number.	Must	**Built**	Link visible on the source record
FR-TR-02	The system shall let staff record milestones with status, location, note and date and time, shown in time order.	Must	**Built**	check:tracking passes
FR-TR-03	The system shall manage customer web accounts with username, email and permission flags for tracking, POD download, document upload and invoice view.	Should	**Built**	Flags saved and reloaded
FR-TR-04	The system shall let a customer sign in to a portal and see only the customer's own files.	Should	**Planned**	No customer sign-in exists in code
FR-TR-05	The customer portal shall show the customer's documents, arrival notices and invoices.	Could	**Planned**	Depends on FR-TR-04

4.13 Public lookup (FR-PT)

ID	Requirement	Priority	Status	Acceptance
FR-PT-01	The public lookup shall show status and milestones for a file number, booking number, air or ocean waybill, master number or house number, including house rows of a consolidation, without a sign-in.	Must	**Built**	check:public-tracking passes for each key type
FR-PT-02	The public response shall contain tracking fields only and no internal financial fields.	Must	**Built**	Response inspection
FR-PT-03	Each public lookup shall require a solved single-use proof-of-work token and shall be rate limited.	Must	**Built**	Lookup without token refused; replay refused

4.14 Live map (FR-MP)

ID	Requirement	Priority	Status	Acceptance
FR-MP-01	The system shall show branches, terminals and ports on an interactive map of the Philippines.	Should	**Built**	Markers appear for directory entries with coordinates
FR-MP-02	The system shall show active dispatch routes on the map from live dispatch data.	Should	**Built**	Active routes drawn
FR-MP-03	Ship and aircraft queries shall be limited to a box of 20 degrees per side or a 250 nautical mile radius, return at most 1,500 markers, and report feed state (including not configured) instead of sample data.	Should	**Built**	check:tracking limit cases pass; map loads with feed off

4.15 Administration, audit, attachments, support and demonstration data (FR-AD)

ID	Requirement	Priority	Status	Acceptance
FR-AD-01	The system shall let an authorised user create, edit and deactivate users and set roles and company access.	Must	**Built**	Changes enforced at next request
FR-AD-02	The system shall let an authorised user maintain companies and their settings, including the billing-to-ledger mode.	Must	**Built**	Settings saved and applied
FR-AD-03	The system shall write an audit record for every mutating request and record booking events for approval changes, and make both available to authorised roles.	Must	**Built**	Each change has a record
FR-AD-04	The system shall attach files to a record (up to 15 MB per request) and accept sign-in help requests as support tickets.	Should	**Built**	Attachment opens from the record; ticket created
FR-AD-05	The system shall show the status of each integration (disconnected, connected or error); live connectors for barcode, label printing, government filing and carrier APIs are required.	Could	**Partial**	Status shown; live connectors Planned
FR-AD-06	The system shall seed demonstration data that is marked so it can be found, and shall not run the seed against production records.	Should	**Built**	Demo rows carry a marker; seed refuses production use

5. Non-Functional Requirements

5.1 Security (NFR-SE)

ID	Requirement	Priority	Status	Acceptance
NFR-SE-01	Passwords shall be stored only as salted one-way hashes.	Must	**Built**	Database inspection shows no plain passwords
NFR-SE-02	Access and refresh tokens shall be signed and short-lived by configuration; signing secrets shall be required in production.	Must	**Built**	Production start refuses missing secrets
NFR-SE-03	Responses shall carry security headers; the content security policy shall block scripts and fonts from other origins; body sizes shall be limited.	Must	**Built**	Header check on a deployed build
NFR-SE-04	Sign-in and public lookup shall be protected by rate limits, throttling and proof of work (FR-AU-04, FR-AU-05, FR-PT-03).	Must	**Built**	Security checks pass
NFR-SE-05	A user shall never read or write records of a company not assigned to them, whatever identifiers are sent.	Must	**Built**	Cross-company request refused in route tests
NFR-SE-06	Feed credentials and secrets shall never reach the browser or the repository.	Must	**Built**	Source and bundle review
NFR-SE-07	A penetration test and dependency audit shall be completed before go-live.	Must	**Planned**	Signed test report

5.2 Auditability (NFR-AU)

ID	Requirement	Priority	Status	Acceptance
NFR-AU-01	Each audit record shall hold user, company, time, action, entity identifier and payload.	Must	**Built**	Sample change shows all five fields
NFR-AU-02	Each ledger posting shall keep the journal reference, and the bridge audit report shall list every posted reference.	Must	**Built**	Posted item listed in the report

5.3 Performance (NFR-PF)

ID	Requirement	Priority	Status	Acceptance
NFR-PF-01	The concurrency benchmark shall run on each pull request.	Must	**Built**	check:benchmark runs in continuous integration
NFR-PF-02	Every list endpoint shall page its results.	Must	**Partial**	Paging on entity lists; endpoint review to confirm all
NFR-PF-03	On the rehearsal copy: Overview first paint no more than 0.6 s; opening a booking no more than 0.4 s; saving no more than 0.2 s; bridge queue with 200 files about 1.0 s.	Must	**Built**	Measured results recorded in the rehearsal notes

5.4 Availability and recoverability (NFR-AV)

ID	Requirement	Priority	Status	Acceptance
NFR-AV-01	The platform shall expose a health endpoint for hosting probes, and a documented backup and restore procedure shall exist for both database files.	Must	**Partial**	Health endpoint present; backup procedure Planned

5.5 Usability (NFR-US)

ID	Requirement	Priority	Status	Acceptance
NFR-US-01	Every screen shall be usable at 390 px width without horizontal page scrolling.	Should	**Partial**	Screen check at 390 px for each module
NFR-US-02	Screens shall be provided and verified in light and dark appearance.	Should	**Built**	Light and dark screenshots in the design records

5.6 Maintainability and delivery (NFR-MT)

ID	Requirement	Priority	Status	Acceptance
NFR-MT-01	Production schema changes shall be additive and applied by a start-up routine, mirrored in the schema file.	Must	**Built**	check:schema-drift and check:schema-heal pass
NFR-MT-02	A change shall merge only when the web and server checks pass in continuous integration.	Must	**Built**	Pull request checks required
NFR-MT-03	Production shall be deployed only from main, one deployment at a time.	Must	**Built**	Deployment workflow with concurrency group

6. Data Requirements

6.1 Core data entities

Table 6-1. Core entities

Entity group	Main content
Users and companies	User (role, company list), refresh token, password history, security event, company settings
Freight	Shipment file (mode, direction, parties, ports, carrier, containers, dimensions, master and house numbers, lines, approval fields), booking events, quote, commodity
Operations	Vehicle, cartage order, driver, fleet vehicle, dispatch route
Rates	Carrier, port, airline, warehouse, party, billing code, tariff, charge package and lines
Billing and bridge	Invoice and invoice sequence, bridge item, check disbursement
Ledger (separate database)	Accounts, group codes, subsidiary groups, banks, suppliers, signatories, vouchers, journal entries and lines, fiscal periods
Tracking	Tracking item with milestones; web account
Common	Attachment, integration status, audit log, support ticket

6.2 Data rules

ID	Rule	Priority	Status	Acceptance
DR-01	Every operational row shall carry a company code set by the server from the signed-in user.	Must	**Built**	Client-sent company value ignored
DR-02	Money shall use exact decimal handling for invoice and ledger totals.	Must	**Built**	check:invoice passes
DR-03	A posted journal shall have equal debit and credit totals.	Must	**Built**	check:bridge passes
DR-04	An issued reference number shall not be reused.	Must	**Built**	Unique sequence checks
DR-05	Deletion shall be limited to roles allowed by the route and shall leave an audit record.	Must	**Built**	Delete route refuses non-permitted roles
DR-06	Attachment content shall move to a separate file store before attachment volume grows.	Could	**Planned**	Attachments served from a file store

7. Other Requirements

7.1 Financial rules

- VAT is 12 percent on taxable billing lines (FR-SH-06).
- Ledger postings follow the bridge code mapping; an unmapped code is refused, not guessed (FR-BR-03).
- The system does not reopen a closed period; reopening is an owner decision made outside the system (FR-FS-04).
- Posted invoices are corrected by reversal and re-issue, not edited (FR-BL-06).

7.2 Records

- Freight and accounting records are the company's records; the platform retrieves them by file number, invoice number and journal reference.

7.3 Operations and support

- Support requests from the sign-in screen are recorded as tickets for administrators (FR-AD-04).
- Maintenance commands that clear operational records are for administrators only and never run against production (C-03).
- A legacy ledger cleanup command takes a verified backup and runs in dry-run mode by default; it has not yet been run on production (Appendix B, item 3).

8. Traceability Matrix

The matrix links each requirement group to its SAD section, to the owner area that confirms it, and to the check or review that verifies it.

Requirement group	SAD section	Confirmed by	Verification
FR-AU-01 to FR-AU-08; NFR-SE-01 to NFR-SE-04	9.4, 11.1	Technical lead	check:security, check:captcha, check:captcha-enforcement
FR-AU-09 to FR-AU-12; NFR-SE-05	2.6, 11.2	Business owner	check:security user policy; route tests
FR-AP-01 to FR-AP-07	3.3, 7.2, 9.1, 11.2	Operations lead	check:opstatus; route tests
FR-OV-01 to FR-OV-02	7.1	Operations lead	Screen review
FR-SH-01 to FR-SH-09	7.2	Operations lead	check:air-booking, check:billing, check:invoice, print review
FR-SH-10 to FR-SH-12	7.3	Operations lead	check:air-booking, check:air-forms
FR-RT-01 to FR-RT-05	7.7	Accounting lead	check:masterdata, check:billing
FR-BL-01 to FR-BL-10	7.8, 9.3	Accounting lead	check:billing, check:invoice, check:flow
FR-VH-01 to FR-VH-03	7.4	Operations lead	Functional review by module
FR-PD-01 to FR-PD-03; FR-FD-01 to FR-FD-04	7.5, 7.6	Operations lead	Functional review by module
FR-BR-01 to FR-BR-07	7.9, 9.2	Accounting lead	check:bridge, qa:accounting
FR-FS-01 to FR-FS-09	7.10, 9.6	Accounting lead	company chart check, qa:accounting
FR-TR-01 to FR-TR-05	7.11	Operations lead	check:tracking; functional review
FR-PT-01 to FR-PT-03	7.11, 9.5	Operations lead	check:public-tracking
FR-MP-01 to FR-MP-03	7.12	Operations lead	check:tracking; screen review with feed on and off
FR-AD-01 to FR-AD-06; IFR-07	7.13, 10.4, 11.5	Business owner	Audit review; route tests; demo rehearsal
IFR-01 to IFR-10	4, 10	Technical lead	Screen check; route tests; header inspection
NFR-AU-01 to NFR-AU-02	11.5, 9.2	Accounting lead	Audit review
NFR-PF-01 to NFR-PF-03; NFR-AV-01	5, 6.5, 12.2	Technical lead	check:benchmark; rehearsal measurements; health check
NFR-MT-01 to NFR-MT-03; C-01	6.2, 6.4	Technical lead	check:schema-drift; pipeline runs
DR-01 to DR-06	8.4	Technical lead	check:invoice, check:bridge, sequence checks

Appendix A. Planned Items Register

Items below are required by this specification but are not implemented in the repository at origin/main.

Requirement	Planned item	Dependency	Note
FR-AP-07	Block self-approval of a file	Decision (risk R-02)	Needs a rule for single-person offices
FR-BL-09	Credit note document	Known gap 1	Until then: void and re-issue
FR-BL-10	Revenue split by service to specific accounts	Known gap 2; accountants' decision	All charges post to the sales account today
FR-TR-04, FR-TR-05	Customer portal sign-in, documents and invoices	Not started	Web account records exist
FR-AD-05, IFR-07	Live device, government and carrier connectors	Connector credentials	Status records exist
NFR-SE-07	Penetration test and dependency audit	Test window before go-live	Self-checks exist
NFR-AV-01	Backup and restore procedure for both databases	Hosting storage decision	Health endpoint exists
DR-06	Attachment file store	Hosting decision	Attachments stored in the database

Appendix B. Known Limitations from the Demo Rehearsal

The rehearsal of 5 November 2026 found the limits below. None loses data. These are limitations of working features, not missing features, unless stated. Items 1 and 2 are the Planned items in Appendix A.

No.	Limitation	Status	Treatment
1	No credit note document; zero or negative invoices refused	**Planned**	Void and re-issue (FR-BL-09)
2	All charges post to the sales account; no split by service	**Planned**	Accountants' decision (FR-BL-10)
3	Legacy ledger cleanup has not been run on production	**Built**	Tested on synthetic ledgers; dry run first, then a verified backup and one reviewed run
4	Tariffs and charge packages are mostly empty; only one lane has a known rate	**Built**	Data entry by the business (A-01); the bridge shows "no tariff" until added
5	Overview income for the month sums files by estimated or actual dates; no booking date exists	**Built**	The widget states the basis
6	VIN decode "Engine" label shows engine layout, not displacement; US-format VINs only; needs internet	**Built**	Do not demonstrate live on a weak connection
7	Operators can edit each other's draft files (shared desk); approved and pending files are locked	**Built**	Accepted by the business; see FR-AP-03
8	Consolidation invoice house number repeats the MAWB number	**Built**	No single house number exists on a consolidation
9	Live map needs internet for tiles and the ship feed token; without the token ships are empty	**Built**	Configuration in host settings (FR-MP-03)
10	Arrival notice email is a preview unless SMTP is configured in the host	**Built**	Configure SMTP before sending
11	Files created through the API, not the form, get a generic file number	**Built**	Use the form for the mode-coded number
12	Not covered by the rehearsal: tracking milestone entry screens, RORO inspection dialog, keyboard focus order, forced server errors	**Built**	Test by hand before each demonstration